Tutorial: Color Check Module

Design: Roland (Ypsos) Contribution: Cricri

Introduction

This module allows you to harmonize the colors of your photographic tiles to avoid color differences between adjacent DDS files.

Step-by-Step Workflow

Step 1: Loading

Open the module

Step 2: Analysis

Ortho4XP will compare the normalization result performed by Color Normalize based on an average derived from 48,753 source JPGs. Color correction is performed per individual RGB channel rather than globally.

ZL/Tile Layers Window (All)

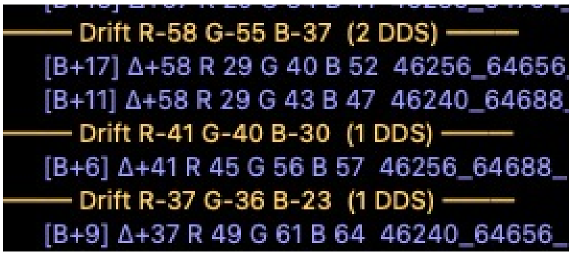
DDS files whose R, G, or B channel deviates from this reference average by more than 5 points (positive or negative) are displayed in the left window. They are grouped by similar drift, and their R, G, B channel values as well as their DDS number and ZL level are indicated.

Search/Filters:

This allows searching by JPG number, ZL, dominant color, as well as by `.comb` zone label (e.g., typing "piste" to find files with protected zones). This is a major productivity boost.

To correct the colorimetry or sharpness, select a DDS per group.

Decision Support: You will see a numeric value displayed called ΔE.



What is it?

It is the color difference between your two sources.

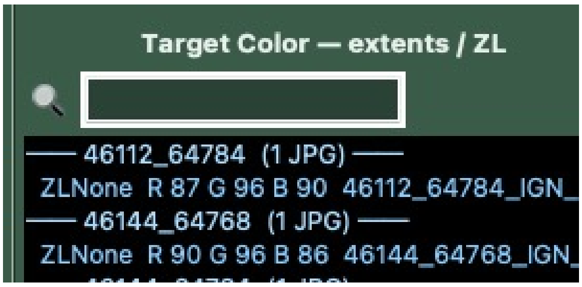
How to use it:

If the ΔE is low: A simple basic color adjustment is often enough to hide the seam.

If the ΔE is high: The difference is too marked. It is then preferable to generate a `.comb` mask (via the dedicated tool) to create a clean transition rather than attempting a complex manual adjustment.

"Target Color -- extends/ZL" Window

DDS files complying with this average value are displayed in the right window with their detailed information.

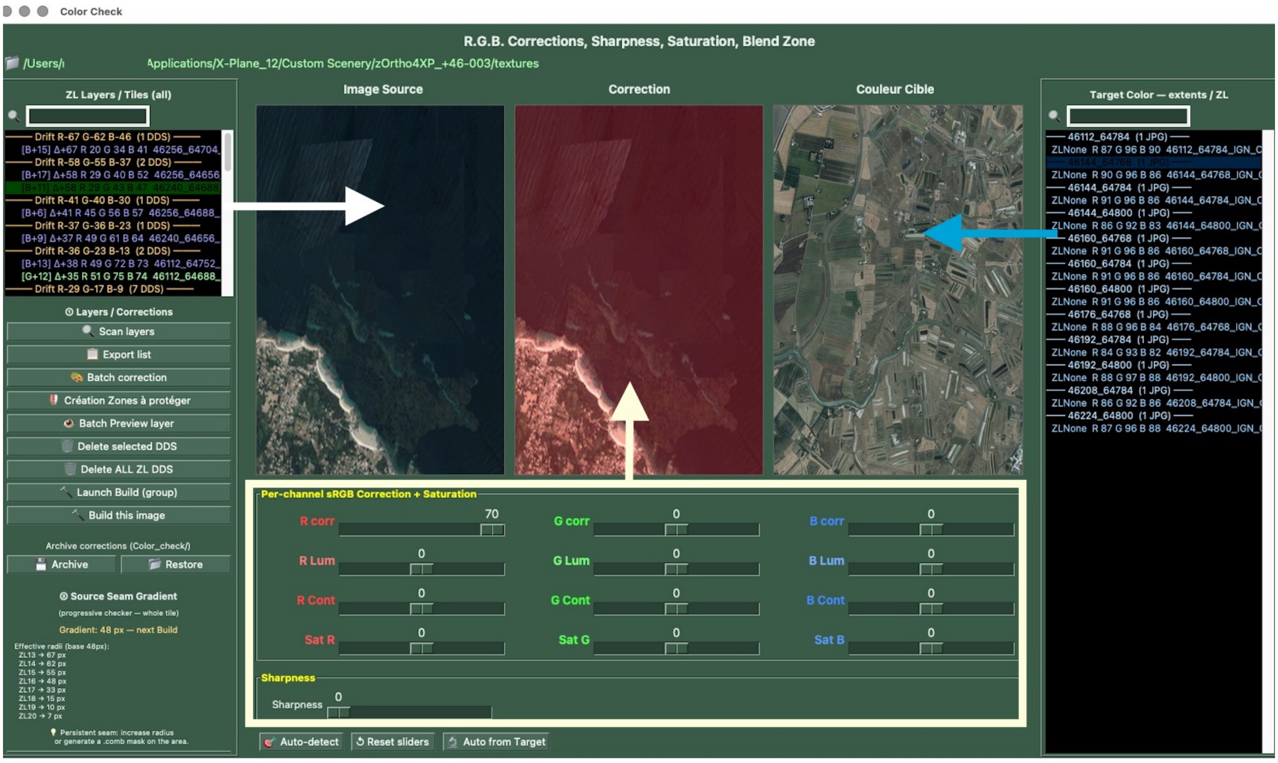


The Windows and Their Display

Window 1 (Left): Displays the selected DDS files.

Window 2 (Center): Displays the corrections made by the sliders or bottom buttons per channel and per R, G, B layer.

Window 3 (Right): Displays the DDS files selected in the right window.



Extract colorimetric data from your textures.

Click the [Scan layers] button

Step 3: Adjustment

After selecting a group DDS from the left panel, 3 correction methods are available.

1. Target Color: allows harmonizing a group relative to another manually chosen JPG.

In the right panel, select a reference DDS and click the "Auto from target" button

2. By Auto-detect: The tool attempts to correct as close as possible to the reference average, based on the European reference average (48,753 JPGs).

In the left panel, select a DDS to correct and click the "Auto-detect" button

3. Central Sliders (sRGB, Saturation): They allow harmonizing the color of DDS files per channel and per layer. In the left panel, select a DDS to correct

Previews and Advanced Previews

👁 Layer Preview: Evaluates at a glance the impact of corrections on an entire ZL layer in the form of thumbnails before applying them permanently.

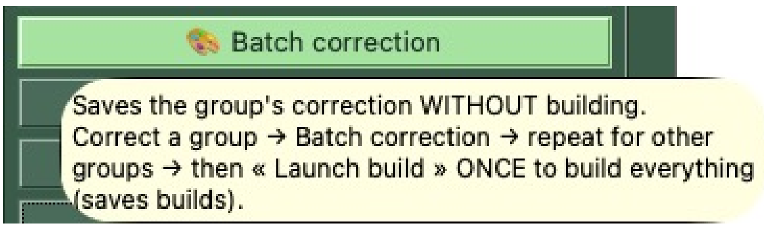
A double-click on a ZL group in the left list directly opens this batch preview.

Delete select: This action deletes only the selected files and avoids Ortho4XP having to recalculate all files.

Delete ALL ZL DDS: This is a destructive action that forces Ortho4XP to recalculate the files, to be used only in case of real necessity to avoid unnecessarily consuming processing time.

Step 4: Corrections and DDS Reconstruction.

Method 1) Once corrections are made on a DDS and are applicable to the whole group, you can click the "Batch correction" button; your settings for this group are saved. You can now go to another group and perform corrections. This method avoids having to wait for the build reconstruction to finish.

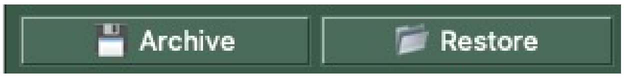


Once corrections are applied either to 1 group or to all groups, click the "Launch construction (group)" button.

The tool will delete all selected DDS files in the current tile, reload the corresponding JPGs, and modify only the selected JPGs of the group. Corrections will be applied and the DDS files will be rebuilt. Under no circumstances are the DDS files in the tile nor the source JPGs corrected.

Method 2) Correct only a single DDS. Once corrections are made on a DDS, you can click the "Build this image" button

Archive and Restore: They allow saving the global corrections of a tile in a `Color_check/` folder at the root of Ortho4XP, which is crucial for project management.

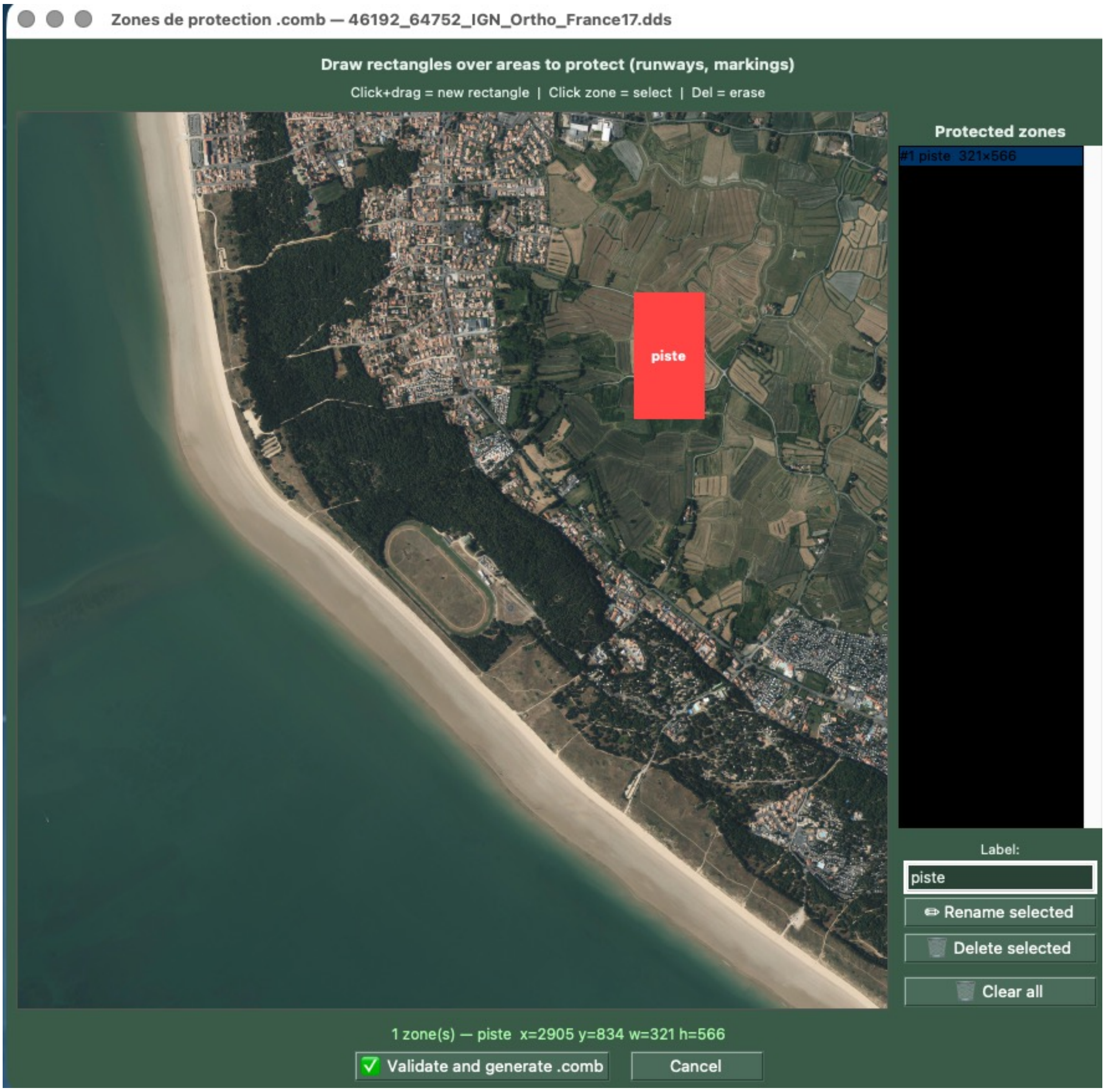


Advanced Functions and .comb Button Management

To go further in refining your textures and optimizing seams, and to isolate zones to protect, the module incorporates advanced tools grouped below:

Creation of Zones to Protect: Allows associating a `.comb` file with a DDS texture to record specific corrections and geometric protection zones.

Zone to Protect Editor: Accessible when creating a `.comb`, it allows you to draw protection rectangles directly with the mouse on runways, markings, or fine details.



Generate seam .comb: Automatically detects the seam line in the selected DDS and generates a geometric protection mask (width based on the current radius) so that the colorimetric correction does not alter the transition zone between two sources.

Tooltip: "Automatically protects seam lines between DDS to avoid visible boundaries (automatically applied during Build)."

Gradient Preview (before construction): If automatic detection and correction of the seam line in the selected DDS is not satisfactory with the "Generate seam .comb" option, it will be possible to proceed with manual adjustment.

Click the "Gradient preview (before construction)" button

The width of the orange bands delineates the blending zone between the two JPEGs. In this zone, pixels blend according to a wavy random sand-grain effect, which fades from the center outward.

For low ZLs (13-16): If your radii are too small, the blend will be too abrupt. Increase them for a more natural gradient.

For high ZLs (18+): Be careful, overly wide radii can alter the sharpness and details of the image. The module will warn you if your settings risk degrading visual sharpness.